

Meeting I: Motivation for Logicians

Topoi: The Categorical Analysis of Logic

Randall David Rickel

May 12, 2026

Goals for today

By the end of this meeting, we should understand that:

- ▶ topos theory is a good subfield for beginning study in categorical logic
- ▶ there are established connections between topos theory and set theory that provide conceptual clarification
- ▶ there are active research questions in set theory that topos theory may help with

Categorical Logic Overview

What is categorical logic?

Categorical logic

The application of tools and concepts from **category theory** to **logic**

Category theory

The general study of mathematical structures through objects and how they relate to other objects. More to come!

Topos theory is a **subfield** of categorical logic. Other topics include:

- ▶ Type theory (or homotopy type theory)
- ▶ Categorical semantics
- ▶ Universal logic
- ▶ Many others

What is topos theory?

Rough definition of topos

A topos is a special category that behaves nicely like many things. For our purposes, intuitionistic formal systems.

We will primarily be focusing on **elementary topos**, which have many nice applications to logic and set theory.

The Elephant

Topoi are often explained with the parable of the blind men and the elephant.

But why are we studying topos theory?

- ▶ Resources for topos theory are friendly
 - *Topoi* by Goldblatt is especially strong
 - ▶ long introduction to prior category theory
 - ▶ scoped to just applications in logic
- ▶ Most of us are interested in set theory, and there are many applications
 - established connections
 - ▶ forcing
 - ▶ souslin's hypothesis
 - conceptual clarification
 - ▶ algebraic set theory
 - ▶ geometric logic
 - ▶ classifying topoi
 - ▶ material/structural set theory
 - active research areas
 - ▶ localic descriptive set theory
 - ▶ monoidal symmetric models
 - ▶ class forcing

Established Connections

Forcing in topoi

Independence proofs through forcing in topoi have existed since around 1970, when Lawvere announced joint work with Tierney.

Sheaf Theory and the Continuum Hypothesis by Tierney (1972)

A few quotes of interest:

- ▶ "... it becomes possible to pass back and forth between models of CS [categorical set theory] and models of set theory [ZF]"
 - There is a strong relationship between set theory formalized in topoi and standard ZF.
- ▶ "... it seems unlikely that these methods, using partially ordered sets, will yield many interesting new results in this area..."
 - This prediction is largely true with this treatment of classical forcing clarifying, but not replacing, existing methods.
- ▶ Tierney goes on to predict this method would prove useful in ways to be discussed.

Forcing continued

We will not cover forcing in topoi this summer. However, we will cover some requisite topics such as:

- ▶ Algebra of Subobjects (Chapter 7)
- ▶ Intuitionism and Kripke Semantics (Chapter 8)
- ▶ Set Concepts and Validity (Chapter 10)
- ▶ Elementary Truth (Chapter 11)
- ▶ Categorical Set Theory (Chapter 12)
- ▶ Local Truth (Chapter 14)
- ▶ Adjoints and Quantifiers (Chapter 15)
- ▶ Logical Geometry (Chapter 16)

Hopefully we continue next semester and can cover forcing from *Sheaves in Geometry and Logic* by MacLane and Moerdijk.

Suslin's Hypothesis

Bunge proved the consistency of the negation of Suslin's hypothesis with CS.

Topos Theory and Souslin's Hypothesis by Marta C. Bunge (1974)

The result of this paper is in the lineage of Tierney and Lawvere's work. However, there are many interesting aspects of this paper:

- ▶ Bunge defines the notion of trees and Suslin trees entirely within the language of category/topos theory
- ▶ She uses a sheaf topos to show results about the global tree which prove facts needed for the argument
- ▶ Finally, she builds the actual forcing poset from finite trees

Does this help with questions about Suslin trees directly? **Probably not.** However, you may be able to apply such constructions to more general forcing contexts. As far as I can tell, this has not been done in the literature.

Suslin's hypothesis continued

Content from the following chapters will be useful for understanding this topic:

- ▶ Algebra of Subobjects (Chapter 7)
- ▶ Elementary Truth (Chapter 11)
- ▶ Categorical Set Theory (Chapter 12)
- ▶ Local Truth (Chapter 14)
- ▶ Adjoints and Quantifiers (Chapter 15)
- ▶ Logical Geometry (Chapter 16)

Geometric Logic

Geometric logic provides nice correspondences between set theory and algebraic geometry.

Rough Definition of Geometric Logic

The fragment of FOL that is robust under sheaf/forcing-style change of universe. The part of logic that behaves continuously under change of topos.

Geometric Logic tells us that some fragments of logic are stable under localization. This is an active area *within* categorical logic (Barr's theorem, Caramello bridges between AG and CL), but the payoff for those interested in more "standard" set theory questions is probably minimal.

Quick note

We will likely cover some of this! Logical Geometry (Chapter 16) covers these topics.

Active Research Areas

Descriptive Set Theory of Locales

Rough definition of DST in Locales

Localic DST is about what parts of DST can be reconstructed from the lattice of opens instead of just points. This separates the DST content that depend on points from the content that only depends on the algebra/topology of opens.

John Isbell opened discussion in localic DST in 1991:

First Steps in Descriptive Set Theory of Locales by John Isbell (1991)

Important quote:

- ▶ "This paper initiates, not a generalization of descriptive set theory, but a reformulation of it admitting "pointless" subsets or *sublocales* of a metrizable space."
 - LDST does not subsume DST, but offers different tools.

LDST Continued

Recent work from Ruiyuan Chen provides a more unified account of this program:

"Borel and analytic sets in locales" by Ruiyuan Chen (2020)

Select quotes:

- ▶ "We systematically develop analogs of basic concepts from classical descriptive set theory in the context of pointless topology."
 - This paper explicitly builds the basics of descriptive set theory in locales
- ▶ "Our starting point is to take the free complete Boolean algebra generated by the frame $O(X)$ of opens to be the ' ∞ -Borel sets' in a locale X ."
 - This is what replaces the Borel subsets of a Polish space.

In the later chapters (14 in particular with sheaves) teaches us the basics of local/open semantics, which is used as the basis for Chen's work.

Class Forcing

Rough definition of class forcing

Class forcing is applying the independence methods of forcing to **proper classes**.

William Easton used forcing on classes to prove his namesake theorem in 1970, which allowed him to control the continuum function at multiple cardinals simultaneously. A more modern question is asking when class forcing behaves close enough to set forcing.

***Class forcing and topos theory* by David Michael Roberts**

David Roberts states that ordinary forcing is the familiar pattern of sheaves on a small site, while class forcing is on a large site. However, small sheaves on a large site are not automatically a topos. This is still a speculative ceiling.

Reading Group Logistics

Schedule

- ▶ Currently, we are scheduled to run the reading group every **Tuesday at 8 pm eastern, 7pm central.**
 - Thoughts?

Note

The schedule, topic list, and presenter assignments can be viewed on the Google Sheet in the Discord server

Meeting Structure

- ▶ This meeting is **abnormal**
- ▶ Future meetings will have the structure of ~1 hour of lecture and ~1 hour of guided problem solving
 - **Lecture** will cover the content from the prior week's reading
 - **Guided problem solving** will cover selected exercises from the previous problem set, some new ones, and discussion

Readings and Problem Sets

- ▶ Readings will be given at the end of lecture
- ▶ Exercises will be posted on the discord server following the meeting
- ▶ Reading/exercises will be in 3 tiers
 - Minimal required reading for comprehension
 - ▶ ~1 hour of work a week for basic preparation for meetings.
 - Good preparation for meetings.
 - ▶ ~3-5 hours a week
 - Extra preparation with additional, external resources.
 - ▶ ~6-10 hours of work a week to understand content more deeply. May include applications to other topics.

Note

We understand schedules are **busy** and **uneven**. Obviously, this group is voluntary and we would still like for you to show up, even if you do not read or do the exercises.

During the week

- ▶ The categorical logic channel is intended for discussion of content during the week. Mo and I will be active to discuss questions you may have.

Note

We are learning some of this material too, but we will do our best to help!

- ▶ If you need the book, please message me on discord or ask at the end of this presentation.

Reading for Next Lecture

- ▶ Minimal:
 - Review the convention of a function from chapter 2.1
 - ▶ Namely, a function is an ordered triple $\langle A, B, R \rangle$ such that $R \subseteq A \times B$ is an appropriate kind of relation
 - Section 2.3
 - Section 2.5 examples 1–7, 9
- ▶ Good preparation:
 - All of chapter 2, especially section 2.3 and all examples of section 2.5
- ▶ Extra preparation:
 - watch these video lectures:
 - ▶ Topos Institute Introduction
 - ▶ Video Lecture
 - ▶ n-Lab on category theory sections 1 and 4